

Statistics Practice Set S32

Level: Intermediate → Slightly Advanced

Focus: Conditional Probability, Bayes, Addition Rule, Complement, Rate vs Count, and Business Interpretation.

No hints. No duplicate questions.

Q1. Conditional Probability — Customer Support

Channel	Customers	Resolved
Chat	1,500	1,200
Email	1,000	700

Find:

1. Total customers
 2. Total resolved cases
 3. $P(\text{Resolved} \mid \text{Chat})$
 4. $P(\text{Chat} \mid \text{Resolved})$
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Q2. Independent vs Dependent — Marketing

Group	Purchased	Did Not Purchase
Saw Ad	300	1,200
Did Not See Ad	150	850

Find:

1. Purchase rate among customers who saw the ad
2. Purchase rate among customers who did not see the ad
3. Are **ad exposure** and **purchase** independent or dependent?
4. Explain your answer in one sentence.

Q3. Addition Rule — App & Website

Out of **3,000** customers:

- 1,800 used the App
- 1,400 used the Website
- 600 used both

Find:

1. $P(\text{App})$
2. $P(\text{Website})$
3. Number of customers who used **at least one**
4. Number of customers who used **neither**

Q4. Complement — Customer Retention

A company has a **9% churn rate**.

Find:

1. Percentage of customers who **do not churn**
 2. Expected number of non-churned customers out of **8,000**
 3. Is the expected number a guarantee? Explain briefly.
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Q5. Mutually Exclusive vs Overlapping — Transactions

A transaction can be classified into:

- **A:** Transaction amount $< ₹500$
- **B:** Transaction is online
- **C:** Transaction amount $> ₹1,000$

Suppose:

- A and B have some transactions in common.
- B and C also have some transactions in common.

Answer:

1. Are A and B mutually exclusive?
 2. Are B and C mutually exclusive?
 3. What does a non-empty intersection tell you?
 4. What would the intersection look like if two events were mutually exclusive?
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Q6. Bayes — Fraud Detection

Transaction Source	Transactions	Fraud
Online	10,000	300
Offline	5,000	100

Find:

1. Total transactions
 2. Total fraud transactions
 3. $P(\text{Fraud} \mid \text{Online})$
 4. $P(\text{Online} \mid \text{Fraud})$
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Q7. Rate vs Count — Sales Campaign

Campaign	Reached	Purchased
A	10,000	700
B	4,000	360
C	2,000	220

Find:

1. Conversion rate of A
 2. Conversion rate of B
 3. Conversion rate of C
 4. Which campaign has the **highest purchase count**?
 5. Which campaign has the **highest conversion rate**?
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Q8. Analyst Interpretation

An analyst says:

“Campaign A is the best because it generated the highest number of purchases.”

Do you agree or disagree?

Give **one reason using conversion rate**.

Q9. Reverse Probability — Employee Skills

Department	Employees	Python Skilled
Data Analytics	400	240
Finance	600	120
Marketing	1,000	100

Find:

1. Total employees
 2. Total Python-skilled employees
 3. $P(\text{Python} \mid \text{Data Analytics})$
 4. $P(\text{Data Analytics} \mid \text{Python})$
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Q10. Full Business Case — Churn

Plan	Customers	Churned
Basic	5,000	650
Premium	2,500	250
Enterprise	1,000	60

Find:

1. Churn rate for each plan
2. Which plan has the highest churn rate?
3. Can you conclude that this plan **causes** churn? Why or why not?